

Deep Learning

Theoretical Exercises - Lesson 1 - Chapter 2

Exercises on the book "Deep Learning" written by Ian Goodfellow,
Yoshua Bengio, and Aaron Courville.

Exercises and solutions by T. Méndez and G. Schuster

FS 2025

1 Exercises on Linear Algebra

1. Calculate the following matrix products.

(a) $C = AB$ and $D = BA$

$$\text{with } A = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 7 & 0 \\ 0 & 1 \\ 3 & 8 \end{pmatrix}$$

Solution:

Simple matrix multiplication (row times column):

$$C = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} \begin{pmatrix} 7 & 0 \\ 0 & 1 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 7 \cdot 7 + 7 \cdot 0 + 0 \cdot 3 & 7 \cdot 0 + 7 \cdot 1 + 0 \cdot 8 \\ 8 \cdot 7 + 4 \cdot 0 + 2 \cdot 3 & 8 \cdot 0 + 4 \cdot 1 + 2 \cdot 8 \end{pmatrix} = \begin{pmatrix} 49 & 7 \\ 62 & 20 \end{pmatrix}$$

$$D = \begin{pmatrix} 7 & 0 \\ 0 & 1 \\ 3 & 8 \end{pmatrix} \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 7 \cdot 7 + 0 \cdot 8 & 7 \cdot 7 + 0 \cdot 4 & 7 \cdot 0 + 0 \cdot 2 \\ 0 \cdot 7 + 1 \cdot 8 & 0 \cdot 7 + 1 \cdot 4 & 0 \cdot 0 + 1 \cdot 2 \\ 3 \cdot 7 + 8 \cdot 8 & 3 \cdot 7 + 8 \cdot 4 & 3 \cdot 0 + 8 \cdot 2 \end{pmatrix} = \begin{pmatrix} 49 & 49 & 0 \\ 8 & 4 & 2 \\ 85 & 53 & 16 \end{pmatrix}$$

(b) $b = Ax$

$$\text{with } A = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} \text{ and } x = \begin{pmatrix} 7 \\ 3 \\ 10 \end{pmatrix}$$

Solution:

Simple matrix multiplication (row times column):

$$b = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} \begin{pmatrix} 7 \\ 3 \\ 10 \end{pmatrix} = \begin{pmatrix} 7 \cdot 7 + 7 \cdot 3 + 0 \cdot 10 \\ 8 \cdot 7 + 4 \cdot 3 + 2 \cdot 10 \end{pmatrix} = \begin{pmatrix} 70 \\ 88 \end{pmatrix}$$

(c) $\mathbf{c} = \mathbf{h}^T \mathbf{A}$
 with $\mathbf{A} = \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix}$ and $\mathbf{h} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

Solution:

Simple matrix multiplication (row times column):

$$\mathbf{c} = \begin{pmatrix} 2 & 5 \end{pmatrix} \begin{pmatrix} 7 & 7 & 0 \\ 8 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 2 \cdot 7 + 5 \cdot 8 \\ 2 \cdot 7 + 5 \cdot 4 \\ 2 \cdot 0 + 5 \cdot 2 \end{pmatrix} = \begin{pmatrix} 54 & 34 & 10 \end{pmatrix}$$

2. Calculate the inverse of the orthogonal matrix

$$\mathbf{A} = \frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}.$$

and check whether your result is correct. Also find the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$$

using the inverse matrix.

Solution:

The inverse of an orthogonal matrix is equal to its transposed. Thus,

$$\mathbf{A}^{-1} = \mathbf{A}^T = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 \\ -2 & 2 & 1 \\ 1 & 2 & -2 \end{pmatrix}.$$

To check the result, the definition of the inverse matrix can be used:

$$\mathbf{A}^{-1}\mathbf{A} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 \\ -2 & 2 & 1 \\ 1 & 2 & -2 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}.$$

Now the solution $\mathbf{x}$ can be found as

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 \\ -2 & 2 & 1 \\ 1 & 2 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}.$$

3. Calculate the L^p norm of the vector $\mathbf{y} = \begin{pmatrix} 4 & -3 \end{pmatrix}^T$ for $p = 1, 2, 3$.

Solution:

The definition of the L^p norm is

$$\|\mathbf{y}\|_p = \left(\sum_i |y_i|^p \right)^{\frac{1}{p}}.$$

Thus,

$$\|\mathbf{y}\|_1 = |y_1| + |y_2| = |4| + |-3| = 7,$$

$$\|\mathbf{y}\|_2 = \sqrt{|y_1|^2 + |y_2|^2} = \sqrt{|4|^2 + |-3|^2} = 5,$$

and

$$\|\mathbf{y}\|_3 = \sqrt[3]{|y_1|^3 + |y_2|^3} = \sqrt[3]{|4|^3 + |-3|^3} = \sqrt[3]{91} = 4.49794.$$

4. The following overdetermined, linear system of equations $A\mathbf{x} = \mathbf{b}$ is given, where

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{pmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 1.8 \\ 3.3 \\ 4.1 \end{pmatrix}.$$

Since there are more linear independent equations than unknowns, there is no solution. For this reason, the most reasonable solution should be found, by using the Moore-Penrose pseudoinverse

$$A^+ = VD^+U^T,$$

for which a singular value decomposition must be carried out. To perform a singular value decomposition, follow the steps below.

- Determine the eigenvalues λ_1 and λ_2 of $\mathbf{B} = A^T A$, where $\lambda_1 > \lambda_2$.
- Determine the Matrix D as $D = \begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \\ 0 & 0 \end{pmatrix}$
- Determine matrix $V = (\mathbf{v}^{(1)} \quad \mathbf{v}^{(2)})$, where $\mathbf{v}^{(i)}$ are the eigenvectors of $\mathbf{B} = A^T A$. Note that $\mathbf{v}^{(1)}$ is the eigenvector to the larger eigenvalue λ_1 .
- Determine matrix $U = (\mathbf{u}^{(1)} \quad \mathbf{u}^{(2)} \quad \mathbf{u}^{(3)})$, where $\mathbf{u}^{(i)}$ are the eigenvectors of $\mathbf{C} = AA^T$. Note that $\mathbf{u}^{(1)}$ is the eigenvector to the largest eigenvalue and $\mathbf{u}^{(3)}$ is the eigenvector to the smallest eigenvalue.
- Check whether U and V are orthogonal matrices and whether $A = UDV^T$
- Determine the Matrix D^+ .
- Determine the Moore-Penrose pseudoinverse A^+ .
- Determine the most reasonable solution $\mathbf{x}$.



Hint:

Use the calculator or python to determine the eigenvalues and eigenvectors.

Solution:

- Matrix $\mathbf{B}$ is

$$\mathbf{B} = \begin{pmatrix} 3 & 12 \\ 12 & 56 \end{pmatrix}$$

and its eigenvalues are

$$\lambda_1 = 58.5904 \quad \text{and} \quad \lambda_2 = 0.4096.$$

- Matrix $\mathbf{D}$ is

$$\mathbf{D} = \begin{pmatrix} \sqrt{58.5904} & 0 \\ 0 & \sqrt{0.4096} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 7.6544 & 0 \\ 0 & 0.6400 \\ 0 & 0 \end{pmatrix}$$

- The eigenvectors of $\mathbf{B}$ are

$$\mathbf{v}^{(1)} = \begin{pmatrix} -0.2110 \\ -0.9775 \end{pmatrix} \quad \text{and} \quad \mathbf{v}^{(2)} = \begin{pmatrix} -0.9775 \\ 0.2110 \end{pmatrix},$$

whereby

$$\mathbf{V} = \begin{pmatrix} -0.2110 & -0.9775 \\ -0.9775 & 0.2110 \end{pmatrix}.$$

(d) Matrix C is

$$C = \begin{pmatrix} 5 & 9 & 13 \\ 9 & 17 & 25 \\ 13 & 25 & 37 \end{pmatrix},$$

its eigenvalues are

$$\lambda_1 = 58.5904, \quad \lambda_2 = 0.4096, \quad \lambda_3 = 0,$$

and its eigenvectors are

$$\mathbf{u}^{(1)} = \begin{pmatrix} -0.2830 \\ -0.5384 \\ -0.7938 \end{pmatrix}, \quad \mathbf{u}^{(2)} = \begin{pmatrix} -0.8679 \\ -0.2085 \\ 0.4508 \end{pmatrix}, \quad \mathbf{u}^{(3)} = \begin{pmatrix} 0.4082 \\ -0.8165 \\ 0.4082 \end{pmatrix}.$$

Thus

$$U = \begin{pmatrix} -0.2830 & -0.8679 & 0.4082 \\ -0.5384 & -0.2085 & -0.8165 \\ -0.7938 & 0.4508 & 0.4082 \end{pmatrix}.$$

(e) Orthogonality-check of U and V :

$$UU^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I,$$

$$VV^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Check of the singular value decomposition:

$$UDV^T = \begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{pmatrix} = A.$$

(f) Matrix D^+ is

$$D^+ = \begin{pmatrix} \frac{1}{\sqrt{\lambda_1}} & 0 \\ 0 & \frac{1}{\sqrt{\lambda_2}} \\ 0 & 0 \end{pmatrix}^T = \begin{pmatrix} 0.1306 & 0 & 0 \\ 0 & 1.5625 & 0 \end{pmatrix}$$

(g) The Moore-Penrose pseudoinverse is

$$A^+ = VD^+U^T = \begin{pmatrix} 4/3 & 1/3 & -2/3 \\ -1/4 & 0 & 1/4 \end{pmatrix}.$$

(h) The most reasonable solution (least-square) is

$$\mathbf{x} = A^+\mathbf{b} = \begin{pmatrix} 4/3 & 1/3 & -2/3 \\ -1/4 & 0 & 1/4 \end{pmatrix} \begin{pmatrix} 1.8 \\ 3.3 \\ 4.1 \end{pmatrix} = \begin{pmatrix} 0.7667 \\ 0.5750 \end{pmatrix}.$$